



Tracking forest recovery: Early biomass and carbon stock monitoring in the Rohingya Refugee camps, Cox's Bazar, Bangladesh

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ABSTRACT

Rohingya influx in Bangladesh destructed about 3200 ha of the reserved forest of Cox's Bazar South Forest Division. The restoration initiatives were launched in 2019 for 292 ha of degraded forest areas. This study assessed biomass and carbon stock in the restored sites of the Rohingya Refugee camps. Thirty-four timber, fodder, fruits, and fuelwood species were planted and the natural regeneration of 14 native species was recorded in the restored sites. The density and basal area ranges of the restored sites were from 0 to 4600 trees/ha and 0 to 33.47 m²/ha, respectively. The total aboveground and below-ground biomass were 33.34±5.76 t/ha and 2.26±0.32 t/ha, respectively. The total carbon stock in five pools varied from 8.84 to 149.86 (t/ha) with a mean of 93.90±4.85 t/ha, with soil carbon representing the largest pool. The mean carbon pools for above-ground (trees and undergrowth), litter, dead woody debris, root and soil (up to 30 cm depth) were 16.37 t/ha, 0.15 t/ha, 0.11 t/ha, 1.03 t/ha and 76.29 t/ha respectively. The best fit allometric models (adjusted R² > 0.97) for the above-ground biomass and carbon stock in trees were $\ln(Y) = 1.30 + 1.11 \ln(x_1)$ and $\ln(Y) = 0.61 + 1.11 \ln(x_1)$, respectively. The derived allometric models demonstrated high accuracy, providing tools for restoration monitoring. These findings highlight the success of reforestation in enhancing biomass and carbon sequestration, although continued protection of restored areas is essential for sustainable forest management and climate mitigation.

1. Introduction

The Rohingya refugee camps in Bangladesh have significantly altered the tropical forest of the Cox's Bazar South Forest Division. Extensive deforestation occurred to accommodate housing for refugees and to develop road networks and other infrastructure, and collection of fuelwoods for cooking purpose (Hassan et al., 2018; Mahamud et al., 2018). The degraded forest areas were aimed to restore forest ecosystem, reduce soil erosion and improve the soil quality. Different national and international non-governmental organizations were involved with those activities. This restoration activities were intensified in 2019 with the construction of different slope management interventions (earthen and bamboo terraces) and plantation of native fast to slow growing tree species. Under these initiatives, 292 ha of degraded forest land was brought under plantation (UNHCR and IUCN, 2018).

Establishment of plantation in degraded sites is a challenge. On the other hand, protecting these newly planted areas is more challenging due to frequent changes in land uses by the authorities-such as new

settlements, roads, and utilities-and the frequent modifications done by the Rohingya refugees including planting broom grass, vegetable gardens, horticultural trees and play grounds, etc. Forest restoration efforts aim to counteract the harmful effects of deforestation and degradation by restoring forest cover, structure, and ecological functions. These efforts can boost biodiversity by creating habitats for a variety of species, including those previously lost due to past disturbances (Shimamoto et al., 2018). Nevertheless, these restoration activities are expected to improve soil quality and support the biodiversity of both soil micro- and macro-organisms (Contos et al., 2024). Additionally, restored forests play a vital role in carbon storage. As trees grow, they absorb carbon from the atmosphere, storing it in their biomass and the soil (Islam et al., 2023). This process helps mitigate climate change by lowering greenhouse gas concentrations in the atmosphere (Pache et al., 2021). Enhanced forest structure and function also support ecosystem services like water regulation and soil stabilization (Wang et al., 2023). While forest restoration positively influences biodiversity and carbon storage, its effectiveness depends on the types of species planted, the

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management practices used, and local environmental conditions (Mo et al., 2023).

Restored forests support local livelihoods by providing ecosystem services such as clean water, timber, and non-timber forest products, highlighting their importance in global climate strategies and sustainable development (IUN, 2019). Restoration approaches depend on the level of degradation, residual vegetation, and desired outcomes (Chazdon, 2008). Forest restoration is a difficult endeavor and, in some cases, may even be impossible. The physical environment of a site influences its biotic components (flora and fauna) and vice versa. Soil quality is shaped by a site's soil parameters, climate, and biological activities. Enhanced soil quality influences plant growth, biodiversity, and overall environmental health (Braimoh and Vlek, 2008). Effective restoration involves more than just planting site-specific species; it also requires proper management and protection of the plantation areas (Keenleyside et al., 2012; Liu et al., 2018).

Monitoring of the restored sites in terms of soil quality, biomass and carbon stock is essential for sustainable development and management (Brown, 2002; Cherubin et al., 2016; Mahmood et al., 2021). Thus, the chosen parameters for monitoring must be straightforward, which are capable of capturing the complex characteristics of soil. Baseline data are also necessary to track soil quality over time (Pierzynski et al., 2005; Gibbs et al., 2007; Bünenmann et al., 2018). Assessing biomass and carbon stock is crucial for monitor the progress or stability of restored forest areas and helps to evaluate the recovery of the ecosystem, carbon sequestration, and the overall role of restoration efforts in combating climate change. Forest ecosystems act as major carbon reservoirs, storing substantial amounts of carbon in both aboveground and below-ground biomass, as well as in soils (IPCC, 2014). The biomass and carbon stock in trees can be estimated by using allometric models. However, the use of allometric models is a crucial decision that can influence the result (Mahmood et al. 2019a). The development and use

of site-specific allometric models are critical for accurate biomass and carbon stock estimation, as they account for local ecological variations that significantly influence biomass distribution and carbon storage (Ketterings et al., 2001; Mahmood et al., 2019b). Different species, growth conditions, and climatic factors across various regions can impact tree dimensions and biomass allocation patterns, making generalized models often unreliable for specific sites (Ketterings et al., 2001; Chave et al., 2005). Site-specific models thus enhance precision in estimating above-ground biomass and carbon stocks, which is essential for climate monitoring, carbon sequestration assessments, and sustainable forest management (Henry et al., 2011; Mahmood et al., 2016; Basuki et al., 2009). Without localized models, carbon estimates risk inaccuracies that could hinder effective policy and conservation strategies (Segura and Kanninen, 2005). This study aimed to: i) estimate the above- and below-ground biomass of the restored sites, ii) assess carbon stocks across five different pools, and iii) develop allometric models for biomass and carbon stock assessment for the early stage of restoration. Effective biomass and carbon stock assessments are essential for understanding the forest restoration efforts, contribution to global climate mitigation strategies, and guiding the sustainable management interventions.

2. Materials and methods

2.1. Description of the study areas

The study sites are situated within the Ukhiya, Nayapara, Whykhong, and Teknaf forest ranges of the Cox's Bazar South Forest Division of Bangladesh. These areas encompass 8881.2 hectares in Whykhong, 7858.3 hectares in Teknaf, and 5108.2 hectares in the Ukhiya forest range (Hossain and Haider, 2020). The geographical location of the refugee camps was in between Ukhiya The latitude of the study area lies

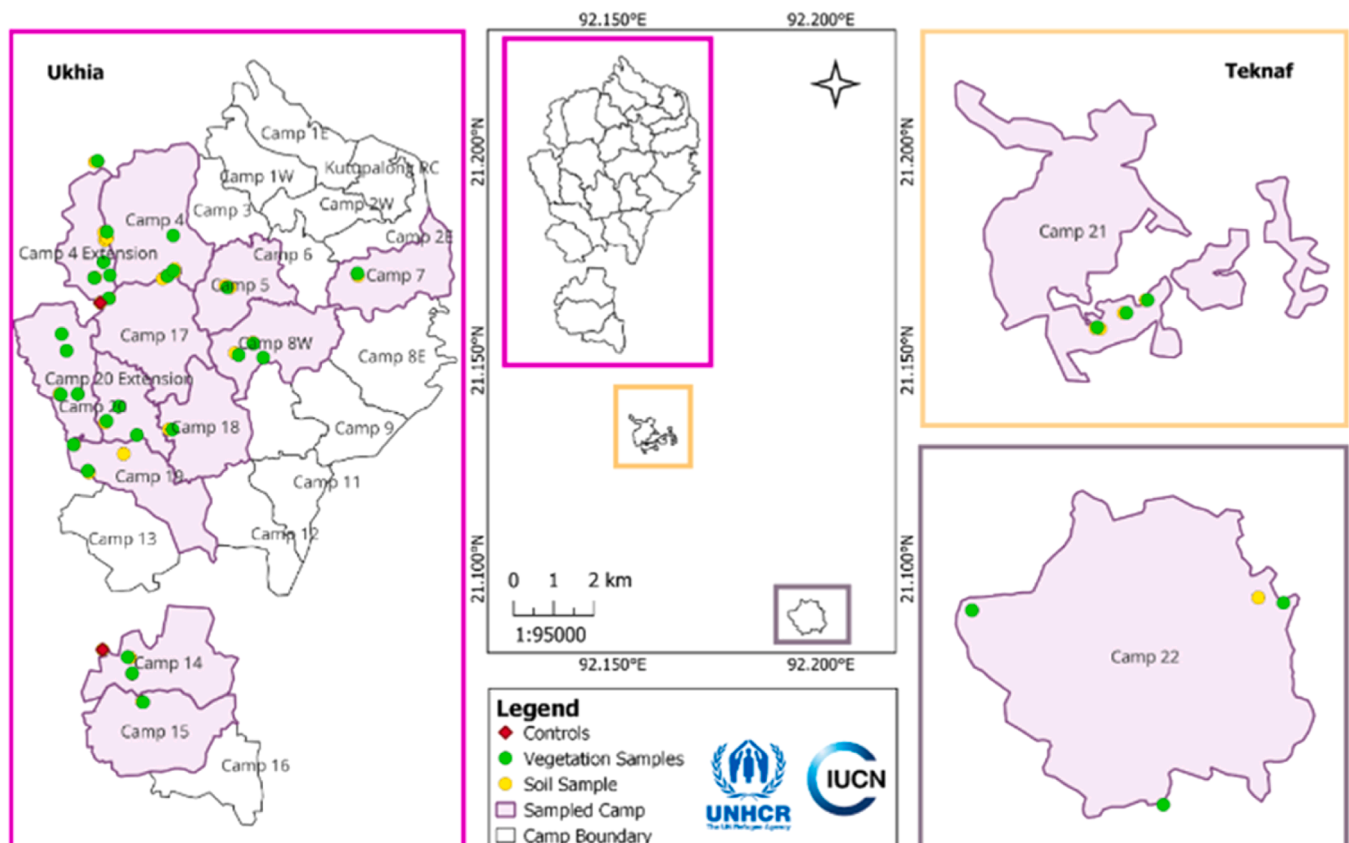


Fig. 1. Location of the Rohingya Refugee camps and field location of the sample plots in the in the Cox's Bazar South Forest Division.

between 21°10'24'' N to 21°20'23''N, and the longitude lies between 92°15'7'' to 92°20'15'' E (Fig. 1). The region experiences a mean annual temperature of 25 °C and average yearly rainfall is approximately 3524 mm, with the lowest recorded rainfall of 4.1 mm in January and the highest, 924.6 mm, occurring in July. The study site consists mainly of low hills of the Dupi Tila formation, with tertiary sedimentary rocks made up of late Miocene and Pliocene unconsolidated sands, clays, silts, and some gravel. A mottled zone often lies above a Plinthite layer. Soils are derived chiefly from sandstone, with quartz, feldspars, and mica in the sand fraction (Huizing, 1971) and textures ranging from loam to sandy clay loam (Zaman et al., 2010; Mahmood et al., 2021). Historically, this area was covered by tropical evergreen forests dominated by *Dipterocarpus* species (Zaman et al., 2010). Over time, the natural forests became degraded, and by 1996, 5539.9 hectares had been converted into plantations of long-rotation tree species (Revilla et al., 1998). Following the Rohingya influx, 55 medium- to long-rotation timber, wild fruit, and fuelwood species were planted for restoration activities (Hossain and Haider, 2020).

2.2. Biomass and carbon stock assessment

2.2.1. Sampling

The restoration activities were conducted in the Rohingya refugee camps in the form of block plantation with variable area coverage. The size of blocks was taking into consideration during the selection process. Blocks having area coverage higher than 0.5 ha (BFD, 2017) considered for the present study. Thus it gave 86 blocks from 14 Rohingya camps. The intention was to cover all camps under-sample. All the blocks of Camp 7, 18, and 19 were selected. Five blocks from camp 4 Ext. and four blocks from camp 20 Ext. were selected. While, three blocks from each camp of 8 W, 19, 20, 21 and 22; two blocks from each camp 4, 14, and 17; one block from each camp of 5 and 15 were selected randomly using strict randomization procedure. Finally, 34 blocks were selected for sample collection. Two denuded sites from Camp 14, and 17 were selected purposively as control (denuded hills) (Table 1, (Fig. 1). The present study established sample plots (10 m x 10 m) in the selected blocks. The typical sample plot (10 m x 10 m) contained four nested plots (1 m x 1 m) at four corners for litter, dead woody debris, and undergrowth samples. The pit (0.5 m x 0.5 m x 0.5 m) layout procedure for estimating the root biomass was very straightforward. The center of the pit was established at the intersection point of two diagonals of the main plot without bias (Fig. 2). *Field measurements*

All trees inside the sample plots were identified using taxonomic

Table 1

Selection of plantation blocks, control sites, vegetable gardens and riparian plantation from the Rohingya Refugee camps.

Camp number	Number of blocks plantation	Selection of sample sites	
		Number of selected blocks	Control (denuded hills)
4	8	2	
4 Extension	11	5	
5	2	1	
7	1	1	
8 W	4	3	
13	1	Nil	
14	4	2	Control
15	5	1	
17	7	2	Control
18	1	1	
19	3	3	
20	8	3	
20 Extension	23	4	
21	4	3	
22	4	3	
Total	86	34	2

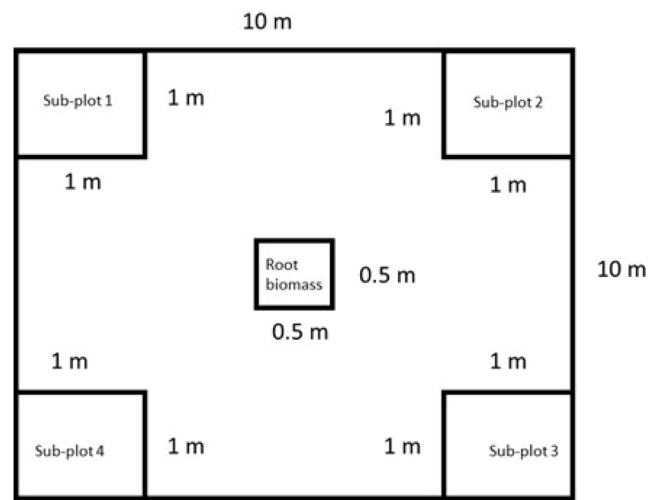


Fig. 2. Sample plot design for the biomass and carbon estimation.

keys; diameter tape was used for DBH measurement. The TruPulse 200 Laser Range Finder was used to measure tree height. All this information about individual trees was recorded for further use. The relative dominance of the tree species was estimated using Eq. (1). Litter, dead woody debris and undergrowth samples were collected from the four nested plots (1 m x 1 m) of each main plot and weighted in the field using a laboratory portable balance. One pit (0.5 m x 0.5 m x 0.5 m) at the center of each plot was excavated. All the soil of the pit was sieved through 2 mm mesh to get the roots. The collected roots for each pit were weighted in the field. Each of the five sub-samples (> 250 g) for litter, dead woody debris, undergrowth and roots were taken and brought back to the laboratory to get fresh to oven-dry at 80 °C weight conversion factor. The oven-dried samples were crushed, sieved and stored for further analysis (Allen, 1989).

$$\text{Relative dominance (\%)} = \frac{\text{Basal area of specific tree species (m}^2\text{/ha)}}{\text{Total basal area of all trees (m}^2\text{/ha)}} \times 100 \quad (1)$$

2.2.2. Soil bulk density and carbon

Soil bulk density was measured for two layers using stainless steel cores (BFD, 2016). The bulk density was calculated according to the following Eqs. (2). Organic carbon content in soil, litter, dead woody debris, undergrowth and fine roots was measured using loss on ignition at 450 °C according to the Eq. (3). Finally organic carbon in samples were estimated from the 50 % of loss on ignition (Allen 1989).

$$\text{Bulk density } \left(\frac{\text{g}}{\text{m}^3} \right) = \frac{\text{Oven - dry weight (g) at } 105^\circ\text{C}}{\text{sample volume (cm}^3\text{)}} \quad (2)$$

$$\text{LOI (\%)} = \frac{\text{Loss of weight after ignition (g)}}{\text{Initial oven-dry weight (g)}} \times 100 \quad (3)$$

2.2.3. Biomass estimation of the sample plots

Species-specific and common allometric biomass models for the hill zone were adopted to estimate the above-ground biomass for trees inside the sample plots (GoB, 2019; Mahmood et al., 2019a). The below-ground root biomass was measured using the pit method (Islam et al., 2023). Litter, dead woody debris, undergrowth and root biomass were estimated based on their fresh weight and the respective conversion factors for oven-dried biomass using Eq. (4) (Mahmood, 2004). Finally, above- and below-ground biomass of the sample plots were expressed in t/ha.

$$\text{Ovendry biomass (kg)} = \text{Fresh weight (kg)} \times \text{conversion factor} \quad (4)$$

2.2.4. Carbon stock assessment

Carbon stock in litter, dead woody debris, undergrowth and roots was estimated from the plot biomass values and carbon concentration in respective components. At the same time, carbon stock in trees was considered to be 50 % of the biomass, according to GoB (2019). Soil organic carbon stock up to 30 cm depth was estimated using soil bulk density and organic carbon concentration. The equations were used to calculate the carbon stock in five pools of the study area has been presented in Table 2.

2.2.5. Allometric models for the monitoring of the aboveground biomass and carbon stock

Allometric models for biomass and carbon stock in above-ground components of a forested ecosystem were developed. Here, plot basis density variables (number of trees and basal area) and plot wise average wood density of the available species were used as independent variables. Wood density values were taken from the database of Sattar (1981). In contrast plot-wise above-ground biomass and carbon stock were the dependent variables. Twenty-one equations (Box 1) were tested to develop the allometric models. At first, all the equations were tested only for the adjusted R² values. The equations with adjusted R² values greater than 0.97 were included for the final model selection. The best fitted models were selected based on CV (Coefficient of Variation), MSE (Mean Squared Error), Rmse (Root Mean Squared Error), F (Fisher's) value, AICc (Corrected Akaike Information Criterion), CF (Correction Factor) and the FI (Furnival index) according to Sprugel (1983), Sileshi (2014) and Mahmood et al. (2019b).

Box 1	
$Y = a + bx_1$	$\ln Y = a + b \ln (x_1, x_2)$
$Y = a + bx_1 + cx_2$	$\ln Y = a + b \ln (x_1, x_3)$
$Y = a + bx_1 + cx_2 + dx_3$	$\ln Y = a + b \ln (x_2, x_3)$
$\ln Y = a + b \ln x_1$	$\sqrt{Y} = a + b \sqrt{x_1}$
$\ln Y = a + b \ln x_1 + c \ln x_2$	$\sqrt{Y} = a + b \sqrt{x_1} + c \sqrt{x_2}$
$\ln Y = a + b \ln x_1 + c \ln x_2 + d \ln x_3$	$\sqrt{Y} = a + b \sqrt{x_1} + c \sqrt{x_2} + d \sqrt{x_3}$
$Y = a + b (x_1, x_2, x_3)$	$\sqrt{Y} = a + b \sqrt{(x_1, x_2, x_3)}$
$Y = a + b (x_1, x_2)$	$\sqrt{Y} = a + b \sqrt{(x_1, x_2)}$
$Y = a + b (x_1, x_3)$	$\sqrt{Y} = a + b \sqrt{(x_1, x_3)}$
$Y = a + b (x_2, x_3)$	$\sqrt{Y} = a + b \sqrt{(x_2, x_3)}$
$\ln Y = a + b \ln (x_1, x_2, x_3)$	Here, Independent variable x ₁ = Basal area, x ₂ = Tree density, x ₃ = wood density

Table 2
Equations for estimating the five pools of carbon in the study area.

Pool type	Equations
Pool 1: Live above-ground	CAGB (t/ha) = AGB of trees (t/ha) x carbon concentration + Biomass of undergrowth (t/ha) x carbon concentration
Pool 2: Litter	CL (t/ha) = LB (t/ha) x carbon concentration
Pool 3: Dead woody debris	CDWD (t/ha) = DWDB (t/ha) x carbon concentration
Pool 4: Below-ground (Fine root)	CBGB (t/ha) = BGB for roots (t/ha) x carbon concentration
Pool 5: Soil organic carbon	SOC (t/ha) = SB (t/ha) x carbon concentration
Total Pool:	Total carbon stock (t/ha) = Pool 1 + Pool 2 + Pool 3 + Pool 4 + Pool 5
AGB = Above-ground biomass	CBGB = Carbon in below-ground biomass
BGB = Below-ground biomass	CDWD = Carbon in dead woody debris
DWDB = Biomass of down woody debris biomass	CL = Carbon in litter
LB = Litter biomass	SOC = Soil organic carbon
SB = Soil biomass	
CAGB = Carbon in above-ground biomass	

2.3. Statistical analysis

Tree density (trees/ha) and basal area (m²/ha) were calculated according to the sample plots. Sample mean was calculated for the carbon concentration in soil, litter, dead woody debris, undergrowth and root. Mean of soil bulk density was calculated for each depth. The adjusted R² values, CV, MSE, Rmse, F value, AICc, CF and the Furnival index for the allometric models were derived using SAS University Edition statistical software.

3. Results

3.1. Vegetation study

Twenty-four planted species of timber, fodder, fruits, and fuelwood and ten naturally regenerated tree species were recorded during this study. The naturally regenerated species were *Alstonia scholaris*, *Chukrasia velutina*, *Dipterocarpus turbinatus*, *Ficus hispida*, *Gmelina arborea*, *Grewia nervosa*, *Pterospermum acerifolium*, *Tectona grandis*, *Terminalia catappa* and *Trema orientalis*. The planted tree species showed higher relative dominance, while the naturally regenerated tree species showed lower. *Albizia procera* had the highest relative dominance (30.84 %), while *Mimusops elengi* had the lowest (0.02 %) (Table 3). However, the sample plots' density and basal area ranged from 0 to 4600 trees/ha and 0 to 33.47 m²/ha, respectively. Comparatively highest density and basal area were observed for camp 15 site 1 and camp 22 site 2 respectively. While, camp 4 site 2, camp 14 site 1, camp 17 site 3 (control), camp 18 site 1 and camp 19 site 3 did not contain any trees (Table 4).

Table 3

Planted and naturally regenerated tree species with their family, local name and relative dominance in the restored sites.

Species name	Family name	Local name	Relative Dominance (%)
<i>Acacia auriculiformis</i>	Fabaceae	Akashmoni	6.74
<i>Albizia lebeck</i>	Fabaceae	Kala koro	3.19
<i>Albizia procera</i>	Fabaceae	Sada koro	30.84
<i>Alstonia scholaris</i>	Apocynaceae	Chatian	0.56
<i>Anisoptera scaphula</i>	Dipterocarpaceae	Boilam	0.05
<i>Artocarpus chaplasha</i>	Moraceae	Chapalish	5.18
<i>Artocarpus heterophyllus</i>	Moraceae	Kanthal	9.03
<i>Bombax ceiba</i>	Bombacaceae	Shimul	5.88
<i>Cassia fistula</i>	Fabaceae	Sonalu	0.07
<i>Chukrasia velutina</i>	Meliaceae	Chikrassi	0.77
<i>Cinnamomum tamala</i>	Lauraceae	Tejapatta	0.05
<i>Dillenia indica</i>	Dilleniaceae	Chalta	0.31
<i>Dipterocarpus turbinatus</i>	Dipterocarpaceae	Telia	0.29
		garjan	
<i>Ficus hispida</i>	Moraceae	Dumur	0.33
<i>Gmelina arborea</i>	Lamiaceae	Gamar	12.94
<i>Grewia nervosa</i>	Malvaceae	Acar gula	0.11
<i>Hopea odorata</i>	Dipterocarpaceae	Telsur	0.73
<i>Magnolia champaca</i>	Magnoliaceae	Champa	0.08
<i>Mangifera indica</i>	Anacardiaceae	Aam	7.60
<i>Mimusops elengi</i>	Sapotaceae	Bokul	0.02
<i>Neolamarckia cadamba</i>	Rubiaceae	Kadam	3.18
<i>Phyllanthus emblica</i>	Phyllanthaceae	Amlaki	1.20
<i>Psidium guajava</i>	Myrtaceae	Peyara	0.07
<i>Pterospermum acerifolium</i>	Malvaceae	Much	0.07
<i>Sterculia villosa</i>	Malvaceae	Udal	0.83
<i>Syzygium grande</i>	Myrtaceae	Jam	1.63
<i>Tabernaemontana divaricata</i>	Apocynaceae	Togor	0.09
<i>Tectona grandis</i>	Lamiaceae	Segun	0.36
<i>Terminalia arjuna</i>	Combretaceae	Arjun	1.68
<i>Terminalia bellirica</i>	Combretaceae	Bahera	1.30
<i>Terminalia catappa</i>	Combretaceae	Badam	0.04
<i>Terminalia chebula</i>	Combretaceae	Haritoki	0.31
<i>Trema orientalis</i>	Cannabaceae	Naricha	4.47

Table 4

Tree density and basal area (m²/ha) in the studied plantation sites during 2024. Sample plots with (*) indicates newly included in the present study.

Camp number	Plots no	Density (trees/ha)	Basal area (m ² /ha)
4	1	400	1.88
4	2	0	0.00
4 Ext.	1	700	3.69
4 Ext.	2	1400	6.83
4 Ext.	3	1000	5.63
4 Ext	4	1600	22.34
4 Ext	5	400	0.99
5	1	800	9.05
7	1	500	7.06
8 W	1	1000	0.80
8 W	2	1300	11.74
8 W	3	1600	16.20
14	1	0	0.00
14	2	1000	6.27
14 Control	3	200	0.13
15	1	4600	11.61
17	1	1100	6.66
17	2	600	4.14
17 Control	3	00	0.00
18	1	0	0.00
19	1	1200	1.00
19	2	200	0.49
19	3	0	0.00
20	1	800	2.86
20	2	1300	2.81
20	3	3000	5.44
20 Ext.	1	600	4.37
20 Ext.	2	2500	13.54
20 Ext.	3	1200	5.00
20 Ext	4	2100	10.09
21	1	700	4.91
21	2	1200	12.72
21	3	1400	8.42
22	1	1200	7.40
22	2	3600	33.47
22	3	2300	15.51

3.2. Soil bulk density and carbon

The mean (1.53±0.02 g/cm³) soil bulk density of two depths (0–15 cm and 15–30 cm) were not varied significantly ($P > 0.05$). Comparatively higher carbon concentration (47.87 %) was detected in dead woody debris, and the lowest (45.14 %) was measured for litter samples (Table 5).

3.3. Above and below-ground biomass

The above-ground biomass of trees for the sample plots ranged from 0 to 143.74 (t/ha). The highest tree biomass (143.74 t/ha) was observed for the sample site 2 of camp 22, and the lowest biomass (0 t/ha) was observed for the sites located in camp 4 site 2, camp 14 site 1, camp 18 site 1, camp 19 site 3, camp 14 control and camp 17 control. Plantations of these sites were lost due to the disturbance of the Rohingya refugees. The litter biomass of the studied sample sites varied with a wide range (0 to 2.47 t/ha). No litter was observed for the twelve sample sites, and the highest litter biomass was observed for sample plot 4 Ext. (1). Like litter, undergrowth, and dead woody debris biomass of the studied sample plots varied with a wide range of 0 to 2.15 t/ha and 0 to 1.09 t/ha,

Table 5

Carbon concentration (%) in litter, dead woody debris undergrowth and roots.

Components	Carbon concentration (%) Mean±SE
Litter	45.14±0.79
Dead woody debris	47.87±0.39
Undergrowth	46.46±0.72
Roots	45.99±0.68

respectively. The mean litter, undergrowth, and dead woody debris of the Rohingya campsites was 0.36±0.08 t/ha, 0.51±0.09 t/ha, and 0.24 ±0.06 t/ha, respectively. Dead woody debris was observed for the eleven sample plots, while five sample plots did not contain undergrowth. The highest undergrowth and dead woody debris were observed for sample plot 4 Ext. (5) (Table 6). The root biomass varied from 0.12 t/ha to 7.48 t/ha with a mean of 2.26±0.32 (t/ha). However, the height below-ground biomass was observed for camp 4 Ext. stie 1 (Table 6).

3.4. Carbon stock

Carbon Pool 1 (Carbon in above-ground trees and undergrowth), Pool 2 (Carbon in litter biomass), Pool 3 (Carbon in dead woody debris), Pool 4 (Carbon in root), and Pool 5 (Carbon in soil up to 30 cm depth) of the studied sample plots varied widely. Total carbon pool varied from 8.84 t/ha to 149.86 t/ha with a mean of 93.95±4.85 t/ha. Lower carbon stocks were observed for the control sites of camp 14 and 17. While the highest carbon stock (149.86 t/ha) was observed for camp 22 site 2 (Table 7). The highest carbon pool for the restored sites were observed

Table 6

Biomass of different components of the restored forest sites.

Camps and sites	Litter (t/ha)	Dead woody debris (t/ha)	Undergrowth (t/ha)	Root (t/ha)	Total above-ground (t/ha)	Above and below-ground (t/ha)
4 (1)	0	0.22	0.58	2.94	9.41	12.35
4 (2)	0.66	0.12	1.08	5.08	1.87	6.95
5 (1)	0.55	0.05	0.12	1.41	48.99	50.41
7 (1)	0	0.13	0.08	3.39	26.09	29.48
14 (1)	0	0.19	0.33	0.41	0.53	0.94
14 (2)	0	0.09	0.05	1.86	30.63	32.50
15 (1)	0.48	0.13	0.18	2.90	61.45	64.35
17 (1)	0.30	0	0.73	4.30	42.35	42.35
17 (2)	0	0	0	1.50	17.85	19.35
18 (1)	0	0	0	0.12	0.00	0.12
19 (1)	0.25	0	0.03	0.83	5.29	6.12
19 (2)	0	0	0.97	1.17	2.49	3.67
19 (3)	0	0	1.00	0.25	1.00	1.25
20 (1)	0.17	0.25	0.42	1.57	13.69	15.26
20 (2)	0.67	0.50	0.52	4.56	13.37	17.93
20 (3)	0.59	0.32	0.46	4.58	17.77	22.35
21 (1)	0.26	0.05	0.30	1.78	26.41	28.19
21 (2)	0.39	0	0.35	1.14	59.63	60.77
21 (3)	0.40	0.16	0.37	2.33	48.65	50.97
22 (1)	0	0	0.75	0.81	35.81	36.62
22 (2)	0	0.85	0.00	0.88	144.60	145.48
22 (3)	0.92	0.48	0.17	1.21	77.47	78.68
14 (°C)	0.29	0.01	0.13	0.39	0.77	1.16
17 (°C)	0	0	0.06	0.78	0.06	0.83
20 Ext (1)	0	0	0.57	0.63	15.43	16.07
20 Ext (2)	0.81	0.91	1.38	2.46	58.07	60.53
20 Ext (3)	0.29	0	1.03	1.06	21.36	22.42
20 Ext (4)	0.21	0.66	1.10	3.93	51.99	55.92
4 Ext (1)	2.47	0.80	0.76	7.48	16.72	24.20
4 Ext (2)	0.40	0.03	0	0.35	41.69	42.04
4 Ext (3)	0.04	0.06	0.92	1.38	26.72	28.10
4 Ext (4)	0.36	0.69	0.85	2.03	127.12	129.15
4 Ext (5)	0.81	1.09	2.15	7.05	6.95	14.00
8 W (1)	0.18	0.23	0.13	1.96	3.61	5.57
8 W (2)	0.57	0.26	0.10	1.61	65.56	67.17
8 W (3)	0.31	0.01	0	5.22	78.99	84.21
Average	0.34 ±0.08	0.23 ±0.05	0.49±0.08	2.26 ±0.32	33.34 ±5.76	35.48 ±5.76

Table 7

Carbon stock (t/ha) in each pool of the studied sites.

Camp and site	Carbon in above-ground tree and undergrowth (t/ha)	Carbon in litter (t/ha)	Carbon in dead woody debris (t/ha)	Carbon in root (t/ha)	Carbon in soil (t/ha) up to 30 cm depth	Total carbon stock (t/ha)
	Pool 1	Pool 2	Pool 3	Pool 4	Pool 5	
4 (1)	4.57	0	0.11	1.34	74.94	80.96
4 (2)	0.5	0.3	0.06	2.33	90.06	93.25
4 Ext (1)	6.69	1.11	0.39	3.42	91.46	103.07
4 Ext (2)	20.63	0.18	0.01	0.16	70.34	91.32
4 Ext (3)	13.28	0.02	0.03	0.63	57.32	71.28
4 Ext (4)	63.01	0.16	0.33	0.93	33.17	97.6
4 Ext (5)	2.46	0.36	0.52	3.22	92.5	99.06
5 (1)	24.2	0.25	0.02	0.65	78.24	103.36
7 (1)	12.98	0	0.06	1.55	81.9	96.49
8 W (1)	1.59	0.08	0.11	0.9	70.82	73.5
8 W (2)	32.36	0.26	0.13	0.74	86.3	119.79
8 W (3)	39.33	0.14	0	2.39	86.44	128.3
14 (1)	0.16	0	0.09	0.19	55.62	56.06
14 (2)	15.27	0	0.04	0.85	73.55	89.71
14	0.23	0.13	0	0.18	39.2	39.74
(Control)						
15 (1)	30.41	0.22	0.06	1.33	55.37	87.39
17 (1)	21	0.13	0	1.97	52.3	75.4
17 (2)	8.93	0	0	0.68	64.78	74.39
17	0.03	0	0	0.36	8.45	8.84
(Control)						
18 (1)	0	0	0	0.05	77.63	77.68
19 (1)	2.52	0.11	0	0.38	99.19	102.2
19 (2)	1.21	0	0	0.54	75.89	77.64
19 (3)	0.46	0	0	0.11	79.94	80.51
20 (1)	6.62	0.08	0.12	0.72	73.07	80.61
20 (2)	6.08	0.3	0.24	2.09	116.19	124.9
20 (3)	8.41	0.27	0.15	2.09	101.57	112.49
20 Ext (1)	7.69	0	0	0.29	99.91	107.89
20 Ext (2)	28.12	0.36	0.44	1.12	107.25	137.29
20 Ext (3)	10.5	0.13	0	0.48	85.33	96.44
20 Ext (4)	25.52	0.09	0.32	1.8	79.92	107.65
21 (1)	13.04	0.12	0.02	0.81	37.56	51.55
21 (2)	29.61	0.17	0	0.52	112.06	142.36
21 (3)	24.03	0.18	0.07	1.06	92.87	118.21
22 (1)	17.88	0	0	0.37	72.92	91.17
22 (2)	71.87	0	0.41	0.4	77.18	149.86
22 (3)	38.03	0.41	0.23	0.55	95.02	134.24

for soil followed by above-ground (trees and undergrowth) (Fig. 3).

3.5. Allometric models for the monitoring of the above-ground biomass and carbon stock in trees of stands

Twenty-one equations were preliminarily tested to derive stand-level models for estimating biomass and carbon stock in trees. This test showed a similar range (0.80 to 0.98) of adjusted R^2 values for the biomass and carbon model (Table 8). Equations with R^2 value > 0.97 were considered for further comparison in both cases. Preliminary, equations 8, 9, 10, 12, 15, 16, and 17 were selected for the biomass and carbon. Adjusted R^2 value, MSE, Rmse, CV, AICc, Furnival Index and

correction factor were computed to select the best fit allometric model for the biomass and carbon stock in trees at stand-level. Equation 8: $\ln(Y) = 1.30 + 1.11 \ln(x_1)$ [adjusted R^2 0.98 with lowest CV (5.22), MSE (0.03), Rmse (0.18), AICc (-105.95), Furnival index (3.77) and correction factor (1.02)] selected as the stand biomass model (Table 9). Similarly, the equation 8: $\ln(Y) = 0.61 + 1.11 \ln(x_1)$ [adjusted R^2 0.98 with lowest CV (6.62), MSE (0.03), Rmse (0.18), AICc (-105.95), Furnival index (3.77) and correction factor (1.02)] selected as best-fit model for the stand carbon stock model (Table 10).

4. Discussion

4.1. Biomass and carbon stock assessment

Most of the sample plots showed lower tree density and some sites were lost their plantation compared to the baseline study of Mahmood et al. (2021). This lower density and loss of plantations in most sample plots indicates higher disturbance and lower protection and maintenance of the plantations. On the contrary, natural regeneration also reported for some sample plots, which gave higher tree density compared to the baseline.

Forest biomass assessment is a critical tool for understanding and managing forest ecosystems, supporting climate change mitigation and carbon sequestration (Pan et al., 2011), promoting sustainable forest management, forest growth and productivity, health, enabling informed decision-making for timber production, conservation, and biodiversity preservation (Brown, 2002), and ensuring accurate monitoring and reporting (Gibbs et al., 2007). Forest biomass information also provides

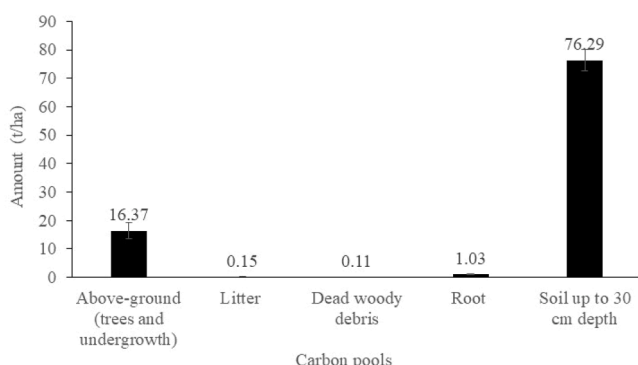


Fig. 3. Average allocation of carbon in different pools of the restored sites.

Table 8

Adjusted regression co-efficient (R^2) of the tested equation. Here, x_1 = Basal area (m^2/ha), x_2 = Tree density (nos/ha), x_3 = wood density (kg/m^3).

Equation number	Equations	Adjusted R^2 value for above-ground biomass of trees	Adjusted R^2 value for carbon in above-ground tree biomass
1	$Y = a + bx_1$	0.96	0.96
2	$Y = a + bx_1 + cx_2$	0.96	0.96
3	$Y = a + bx_1 + cx_2 + dx_3$	0.96	0.96
4	$Y = a + b(x_1, x_2, x_3)$	0.80	0.80
5	$Y = a + b(x_1, x_2)$	0.81	0.81
6	$Y = a + b(x_1, x_3)$	0.94	0.94
7	$Y = a + b(x_2, x_3)$	0.91	0.91
8*	$Ln(Y) = a + bLn(x_1)$	0.98	0.98
9*	$Ln(Y) = a + bLn(x_1) + cLn(x_2)$	0.98	0.98
10*	$Ln(Y) = a + bLn(x_1) + cLn(x_2) + dLn(x_3)$	0.98	0.98
11	$Ln(Y) = a + bLn(x_1, x_2, x_3)$	0.96	0.96
12*	$Ln(Y) = a + bLn(x_1, x_2)$	0.97	0.97
13	$Ln(Y) = a + bLn(x_1, x_3)$	0.96	0.96
14	$Ln(Y) = a + bLn(x_2, x_3)$	0.86	0.86
15*	$\sqrt{Y} = a + b\sqrt{x_1}$	0.97	0.97
16*	$\sqrt{Y} = a + b\sqrt{x_1} + c\sqrt{x_2}$	0.97	0.97
17*	$\sqrt{Y} = a + b\sqrt{x_1} + c\sqrt{x_2} + d\sqrt{x_3}$	0.97	0.97
18	$\sqrt{Y} = a + b\sqrt{(x_1, x_2, x_3)}$	0.86	0.86
19	$\sqrt{Y} = a + b\sqrt{(x_1, x_2)}$	0.87	0.87
20	$\sqrt{Y} = a + b\sqrt{(x_1, x_3)}$	0.94	0.94
21	$\sqrt{Y} = a + b\sqrt{(x_2, x_3)}$	0.90	0.90

essential insights for policy-making and management practices to ensure the health and sustainability of forests. The mean biomass of trees in the present study was only 27 % of the Hill Forest of Bangladesh. Litter, dead woody debris, and undergrowth contribute only 0.05 % of the hill forest (GoB, 2019). This lower biomass of the study sites indicated the large gap between the natural forest of Bangladesh and similar geomorphological settings and climatic conditions. Restored tropical forests generally exhibit lower biomass stocks compared to old-growth tropical forests due to factors like species composition and age (Shimamoto et al., 2018). However, the biomass of the sample plots was negligible during the baseline study of Mahmood et al. (2021). In this regard, the present study has provided substantial information on biomass stocking in the restoring sites of the Rohingya Refugee camps. This information might

help to monitor the forest's health and contribute to decision-making regarding the management and protection of the planted sites.

Forest carbon pools, including biomass, deadwood, litter and soil organic carbon play an important role in carbon sequestration. Assessing these pools helps quantify the amount of carbon stored in forests, which is vital for developing strategies to mitigate climate change (Pan et al., 2011). Forest carbon pools are closely linked to biodiversity and other ecosystem functions and services. Assessing these pools helps in maintaining the integrity of forest ecosystems and forest health (Malhi and Grace, 2000). Quantification and distribution of carbon in different pools in the forest help to adopt appropriate forest management and conservation strategies to enhance carbon sequestration (Ryan et al., 2010). The carbon stock in five pools of the study area was 93.90 ± 4.85 t/ha. Meanwhile, the Hill Forest of Bangladesh contained 1.5 times higher carbon stock (144.42 t/ha) in five pools (GoB, 2019). The total carbon stock of the present study was lower than different forests of Bangladesh (Gibbs et al., 2007; Ullah and Al-Amin, 2012; Khan et al., 2020). Global comparisons of biomass and carbon stock in restored tropical forests highlight significant variability shaped by regional differences in biodiversity, restoration age, and ecosystem management (Shimamoto et al., 2018). For example, the above-ground biomass and carbon stock in different forests of Northeast India ranged between 37.85 - 85.58 t/ha and 18.92 t/ha to 35.13 t/ha, respectively (Majumdar et al., 2016). Similarly, Latin American and Southeast Asian tropical forests report that restored areas generally recover 50–70 % of natural forest biomass within 20–40 years, with mature forests in the Amazon achieving 150–200 tons of carbon per hectare (Romijn et al., 2019). Differences in forest types, climatic conditions, soil quality, age of plantation and management practice may significantly influence the forest biomass and carbon stock among the forests (Keith et al., 2009; Pan et al., 2011). Moreover, the diverse planting schemes and long-term management is important to maximize carbon capture across restored tropical forests worldwide (Majumdar et al., 2016). For more detailed data and specific region comparisons, recent studies provide valuable insights into restoration and carbon stock potential across various tropical regions.

4.2. Allometric models for above-ground biomass and carbon stock assessment

Using the allometric equations, the biomass and carbon stock assessment showed a good correlation with the observed data, with R^2 values generally above 0.80, indicating a reliable relationship between the measured tree dimensions and the estimated biomass (Brown, 1997; Chave et al., 2005). Species, zone and site specific allometric model provides accurate biomass and carbon stock assessment. In case of Bangladesh, fewer species specific allometric models were reported (Mahmood et al., 2016).

The best-fitted models were selected considering the highest

Table 9

Comparison of models for the above-ground biomass in trees. Equation with (*) indicates best fit allometric biomass models. Here, x_1 = Basal area (m^2/ha), x_2 = Tree density (nos/ha) and x_3 = wood density (kg/m^3).

SL no	Equation	Adjusted R2	a	b	c	d	Sa	Sb	Sc	Sd	CV	MSE	Rmse	F value	AICc	CF	FI
8*	$Ln(Y) = a + bLn(x_1)$	0.98	1.30	1.11			0.05	0.03			5.22	0.03	0.18	1684.75	-105.95	1.02	3.77
9	$Ln(Y) = a + bLn(x_1) + cLn(x_2)$	0.98	1.33	1.12	-0.01		1.04	0.11	0.17		5.21	0.03	0.18	813.36	-105.96	1.02	3.84
10	$Ln(Y) = a + bLn(x_1) + cLn(x_2) + dLn(x_3)$	0.98	1.62	1.12	0.00	-0.05	1.56	0.11	0.18	0.19	5.11	0.03	0.18	524.08	-106.03	1.02	3.91
12	$Ln(Y) = a + bLn(x_1, x_2)$	0.97	-2.82	0.69			0.19	0.02			5.15	0.05	0.22	1054.98	-91.75	1.02	4.75
15	$\sqrt{Y} = a + b\sqrt{x_1}$	0.97	-0.29	2.27			0.20	0.07			10.00	0.22	0.47	1019.07	-45.18		4.35
16	$\sqrt{Y} = a + b\sqrt{x_1} + c\sqrt{x_2}$	0.97	-0.25	2.32	0.00		0.42	0.44	0.04		9.70	0.23	0.48	492.22	-45.20		4.43
17	$\sqrt{Y} = a + b\sqrt{x_1} + c\sqrt{x_2} + d\sqrt{x_3}$	0.97	-0.06	2.33	0.00	-0.01	1.08	0.45	0.04	0.04	9.72	0.24	0.49	316.87	-45.24		4.51

Table 10
Comparison of models for the carbon stock in tree biomass. Equation with (*) indicates best fit allometric biomass models. Here, x1= Basal area (m²/ha), x2= Tree density (nos/ha) and x3= wood density (kg/m³).

SL no	Equation	Adjusted R2	a	b	c	d	Sa	Sb	Sc	Sd	CV	MSE	Rmse	F value	AICc	CF	FI
8*	Ln(Y)=a+bLn(x1)	0.98	0.61	1.11			0.05	0.03			6.62	0.03	0.18	1684.75	-105.95	1.02	1.89
9	Ln(Y)=a+bLn(x1) +cLn(x2)	0.98	0.64	1.12	-0.01		1.04	0.11	0.17		6.61	0.03	0.18	813.36	-105.96	1.02	1.92
10	Ln(Y)=a+bLn(x1) +cLn(x2)+dLn(x3)	0.98	0.92	1.12	0.00	-0.05	1.56	0.11	0.18	0.19	6.51	0.03	0.18	524.08	-106.03	1.02	1.95
12	Ln(Y)=a + b Ln(x1 . x2.)	0.97	-3.51	0.69			0.19	0.02			6.55	0.05	0.22	1054.98	-91.75	1.02	2.37
15	√Y = a + b√x1	0.97	-0.20	1.61			0.14	0.05			9.95	0.11	0.33	1019.07	-66.67		2.18
16	√Y = a + b√x1 +c√x2	0.97	-0.17	1.64	0.00		0.30	0.31	0.03		9.65	0.11	0.34	492.22	-66.69		2.21
17	√Y = a + b√x1 +c√x2+d√x3	0.97	-0.04	1.64	0.00	-0.01	0.76	0.31	0.03	0.03	9.66	0.12	0.34	316.87	-66.73		2.25

adjusted R² value and lowest CV, MSE, Rmse, AICc and Furnival index, which was in line with the study of Soares and Schaeffer-Novelli (2005); Mahmood et al. (2015). The derived allometric models highlight the significant potential of reforestation efforts in monitoring the carbon stock and sequestering at the early stages of forest stand development. The variability in biomass and carbon stock in trees in a plot can be attributed to differences in tree density, species composition, and site conditions (IPCC, 2006). The results suggest that for a more accurate assessment of total carbon stock in young stands, it is crucial to use or develop specific models that account for the unique growth patterns and biomass distribution of young trees. These findings underscore the importance of continuous monitoring and data collection to refine carbon stock estimation models for reforestation efforts. The study emphasizes the need for integrating field measurements with model development to improve the accuracy of carbon stock assessments, thereby supporting more effective forest management and carbon sequestration strategies.

5. Conclusion

This study provided a detailed assessment of above and below-ground biomass and carbon stock within reforested areas of the Rohingya refugee camps in Cox’s Bazar, Bangladesh. Thirty-three tree species were planted in the degraded sites as restoration initiatives. However, this restoration activity has successfully enhanced above- and below-ground biomass, with corresponding increases in carbon stocks across multiple carbon pools, including live biomass, litter, deadwood, and soil organic carbon. The developed site-specific allometric models for biomass and carbon estimation (achieving an R² value greater than 0.97) are essential for tracking carbon stock changes over time, refining sequestration estimates, and supporting carbon offset initiatives. Despite challenges like periodic disturbances and land-use modifications by refugee populations, the restoration sites have shown significant carbon storage potential relative to baseline levels, though values remain lower compared to mature tropical forests. These findings highlight the capacity of targeted reforestation within heavily impacted areas to contribute to carbon mitigation and ecosystem stabilization, presenting a replicable framework for similar interventions in tropical regions affected by forced migration and environmental degradation.

CRediT authorship contribution statement

Hossain Mahmood: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Chameli Saha:** Visualization, Validation, Data curation, Conceptualization. **Sanjoy Saha:** Visualization, Validation, Project administration, Investigation, Data curation.

Declaration of competing interest

There are no conflict of interest.

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Data availability

Data will be made available on request.

References

Allen, S.E., 1989. Chemical Analysis of Ecological Materials. Blackwell Scientific Publications, Oxford.

Basuki, T.M., van Laake, P.E., Skidmore, A.K., Hussin, Y.A., 2009. Allometric equations for estimating the above-ground biomass in tropical lowland Dipterocarp forests. For. Ecol. Manag. 257 (8), 1684–1694.

BFD, 2016. Manual For Soil Measurements For the Bangladesh forest Inventory. BangladeshForest Department, and Food and Agricultural Organization of the United Nations, Dhaka, Bangladesh.

BFD, 2017. Definition of Forest for Bangladesh. <http://www.bforest.gov.bd/site/page/eca2a291-b487-4c7a-8ce0-dbe5069524ac/>. Accessed on 18 September 2024.

Braimoh, A.K., Vlek, P.L.G., 2008. Land Use and Soil Resources. Springer Science, pp. 167–200.

Brown, S., 2002. Measuring carbon in forests: current status and future challenges. Environ. Pollut. 116 (3), 363–372.

Brown, S., 1997. Estimating biomass and biomass change of tropical forests: a primer. FAO Forestry Paper 134. FAO.

Bünemann, E.K., Bongiorno, G., Bai, Z., Creamer, R.E., Deyn, G.E., de Goede, R., Flesskens, L., Geissen, V., Kuyper, T.W., Mäder, P., Pulleman, M., Sukkel, W., van Groenigen, J.W., Brussaard, L., 2018. Soil quality – A critical review. Soil Biol. Biochem. 120, 105–125.

Chave, J., Andalo, C., Brown, S., Cairns, M.A., Chambers, J.Q., Eamus, D., Yamakura, T., 2005. Tree allometry and improved estimation of carbon stocks and balance in tropical forests. Oecologia 145 (1), 87–99.

Chazdon, R.L., 2008. Beyond Deforestation: restoring Forests and Ecosystem Services on Degraded Lands. Science 320, 1458–1460. <https://doi.org/10.1126/science.1155365>.

Cherubin, M.R., Karlen, D.L., Franco, A.L.C., Cerri, C.E.P., Tormena, C.A., Cerri, C.C., 2016. A Soil Management Assessment Framework (SMAF) Evaluation of Brazilian Sugarcane Expansion on Soil Quality. Soil Sci. Soc. Am. J. <https://doi.org/10.2136/sssaj2015.09.0328>.

Contos, P., Murphy, N.P., Kayll, Z.J., Morgan, T., Vido, J.J., Decker, O., Gibb, H., 2024. Rewilding soil and litter invertebrates and fungi increases decomposition rates and alters detritivore communities. Ecol. Evol. 14 (3), e11128. <https://doi.org/10.1002/ece3.11128>.

Gibbs, H.K., Brown, S., Niles, J.O., Foley, J.A., 2007. Monitoring and estimating tropical forest carbon stocks: making REDD a reality. Environ. Res. Lett. 2 (4), 045023.

- GoB, 2019. Tree and Forest Resources of Bangladesh: The results of the Bangladesh Forest Inventory Cycle 1. Forest Department, Ministry of Environment, Forest and Climate Change, Government of the People's Republic of Bangladesh, Dhaka, Bangladesh.
- Hassan, M.M., Smith, A.C., Walker, K., Rahman, M.K., 2018. Rohingya Refugee Crisis and Forest Cover Change in Teknaf. Bangladesh. Remote Sens. 10, 689.
- Henry, M., Picard, N., Trotta, C., Manlay, R.J., Valentini, R., Bernoux, M., Saint-André, L., 2011. Estimating tree biomass of sub-Saharan African forests: a review of available allometric equations. Silva Fenn 45 (3), 477–569.
- Hossain, K.M., Haider, R.M., 2020. Assessment Report on Plantation/Reforestation 2019 At Rohingya Camp, Cox's Bazar, Final Draft. IUCN, Bangladesh.
- Huizing, H.G., 1971. A reconnaissance study of the mineralogy of sand fractions from East Pakistan sediments and soils. Geoderma 6, 109–133.
- IPCC, 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories, 4. Agriculture, Forestry, and Other Land Use.
- IPCC, 2014. Climate Change 2014: Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change.
- Islam, S.M., Saha, C., Mahmood, H., 2023. Biomass and carbon stocks in mangrove afforested areas, central coastal areas of Bangladesh. Environmental Challenges 13, 100784. <https://doi.org/10.1016/j.envc.2023.100784>.
- Keenleyside, K.A., Dudley, N., Cairns, S., Hall, C.M., Stolton, S., 2012. Ecological Restoration for Protected Areas: Principles, Guidelines and Best Practices. IUCN, Gland, Switzerland x + 120 pp.
- Keith, H., Mackey, B.G., Lindenmayer, D.B., 2009. Re-evaluation of forest biomass carbon stocks and lessons from the world's most carbon-dense forests. Proc. Natl. Acad. Sci. 106 (28), 11635–11640. <https://doi.org/10.1073/pnas.0901970106>.
- Ketterings, Q.M., Coe, R., van Noordwijk, M., Ambagau, Y., Palm, C.A., 2001. Reducing uncertainty in the use of allometric biomass equations for predicting above-ground tree biomass in mixed secondary forests. For. Ecol. Manag. 146 (1–3), 199–209.
- Khan, M.N.I., Islam, M.R., Rahman, A., Azad, M.S., Mollick, A.S., Kamruzzaman, M., Knohl, A., 2020. Allometric relationships of stand level carbon stocks to basal area, tree height and wood density of nine tree species in Bangladesh. Glob. Ecol. Conserv. 22, e01025.
- Liu, C.L.C., Kuchma, O., Krutovsky, K.V., 2018. Mixed-species versus monocultures in plantation forestry: development, benefits, ecosystem services and perspectives for the future. Glob. Ecol. Conserv. 15, e00419.
- Mahamud, R., Arif, T., Jalal, R., Billah, M., Rahman, L.M., Rahman, T., Iqbal, Z., Henry, M., 2018. Technical Specification For Land Stabilization and Greening the Camp in Cox's Bazar South Forest Division. Bangladesh Forest Department, Dhaka, Bangladesh, p. 52.
- Mahmood, H., 2004. Biomass, Litter Production and Selected Nutrients in *Bruguiera parviflora* (Roxb.) Wight & Arn. Dominated Mangrove Forest Ecosystem At Kuala Selangor, Malaysia. University Putra Malaysia, Seri Kembangan. PhD thesis unpublished.
- Mahmood, H., Saha, C., Paul, N., Deb, S., Abdullah, S.M.R., Tanvir, M.S.S.I., Bashar, A., Roy, S., Rabby, F., Ahmed, S.N., Ali, M.H., 2021. The soil quality of the world's largest Refugee campsites located in the Hill Forest of Bangladesh and the way forward to improve the soil quality. Environ. Chall. 3, 100048. <https://doi.org/10.1016/j.envc.2021.100048>.
- Mahmood, H., Siddique, M.R.H., Islam, S.M.Z., Abdullah, S.M.R., Matieu, H., Iqbal, M.Z., Akhter, M., 2019a. Applicability of semi-destructive method to derive allometric model for estimating aboveground biomass and carbon stock in the Hill zone of Bangladesh. J. For. Res. 31, 1235–1245. <https://doi.org/10.1007/s11676-019-00881-5>.
- Mahmood, H., Siddique, M.R.H., Akhter, M., 2016. A critical review and database of biomass and volume allometric equation for trees and shrubs of Bangladesh. IOP Conf. Ser. Earth Environ. Sci. 39, 012057. <https://doi.org/10.1088/1755-1315/39/1/012057>.
- Mahmood, H., Siddique, M.R.H., Costello, L., Birigazzi, L., Abdullah, S.M.R., Henry, M., Siddiqui, M., Aziz, M., Ali, M., Mamun, A., Forhad, M., Akhter, M., Iqbal, M., Mondol, F., 2019b. Allometric models for estimating biomass, carbon and nutrient stock in the Sal Zone of Bangladesh. iForest 12, 69–75. <https://doi.org/10.3832/ifor2758-011>.
- Mahmood, H., Siddique, M.R.H., Saha, S., Abdullah, S.M.R., 2015. Allometric models for biomass, nutrients and carbon stock in *Excoecaria agallocha* of the Sundarbans. Wetl. Ecol. Manag. 23 (4), 765–774. <https://doi.org/10.1007/s11273-015-9419-1>.
- Majumdar, K., Choudhary, B.K., Datta, B.K., 2016. Aboveground Woody Biomass, Carbon Stocks Potential in Selected Tropical Forest Patches of Tripura, Northeast India. Open J. Ecol. 6 (10). <https://doi.org/10.4236/oje.2016.610057>.
- Malhi, Y., Grace, J., 2000. Tropical forests and atmospheric carbon dioxide. Trends Ecol Evol 15 (8), 332–337.
- Mo, L., Zohner, C.M., Reich, P.B., et al., 2023. Integrated global assessment of the natural forest carbon potential. Nature 624, 92–101. <https://doi.org/10.1038/s41586-023-06723-z>.
- Pache, R.G., Abrudan, I.V., Niță, M.D., 2021. Economic Valuation of Carbon Storage and Sequestration in Retezat National Park, Romania. Forests 12, 43. <https://doi.org/10.3390/f12010043>.
- Pan, Y., Birdsey, R.A., Fang, J., Houghton, R., Kauppi, P.E., Kurz, W.A., Hayes, D., 2011. A large and persistent carbon sink in the world's forests. Sci 333 (6045), 988–993.
- Revilla, J.A.V., Ahmed, I.U., Mabud, A., 1998. Final report: Forest inventory of the Natural Forest and Forest Plantations (Cox's Bazar Forest Division). Forest Department. Ministry of Environment and Forests, People's Republic of Bangladesh, Dhaka.
- Romijn, E., Coppus, R., De Sy, V., Herold, M., Roman-Cuesta, R.M., Verchot, L.L., 2019. Restoration in Latin America and the Caribbean: an Overview of Recent, Ongoing and Planned Restoration Initiatives and Their Potential for Climate Change Mitigation. Forests 10, 510. <https://doi.org/10.3390/f10060510>.
- Ryan, M.G., Harmon, M.E., Birdsey, R.A., Giardina, C.P., Heath, L.S., Houghton, R.A., ... Skog, K.E., 2010. A synthesis of the science on forests and carbon for US forests. Ecological Society of America: Issues In Ecology. 13, 1–16.
- Sattar, M.A., 1981. Some Physical Properties of 116 Bangladeshi timbers. Bulletin no. 7, Wood seasoning Series. Forest Research Institute, Chittagong, Bangladesh, p. 15.
- Segura, M., Kanninen, M., 2005. Allometric models for tree volume and total aboveground biomass in a tropical humid forest in Costa Rica. Biotropica 37 (1), 2–8.
- Shimamoto, C.Y., Padial, A.A., da Rosa, C.M., Marques, M.C.M., 2018. Restoration of ecosystem services in tropical forests: a global meta-analysis. PLoS ONE 13 (12), e0208523. <https://doi.org/10.1371/journal.pone.0208523>.
- Sileshi, G.W., 2014. A critical review of forest biomass estimation models, common mistakes and corrective measures. For. Ecol. Manag. 329, 237–254. <https://doi.org/10.1016/j.foreco.2014.06.026>.
- Soares, M.L.G., Schaeffer-Novelli, Y., 2005. Above-ground biomass of mangrove species. I. Analysis of models. Estuar. Coast. Shelf Sci. 65, 1–18.
- Sprugel, D.G., 1983. Correcting for bias in log transformed allometric equations. Ecology 64 (1), 209–210. <https://doi.org/10.2307/1937343>.
- Ullah, M., Al-Amin, M., 2012. Above-and below-ground carbon stock estimation in a natural forest of Bangladesh. J. For. Sci. 58, 372e379.
- UNHCR, IUCN, 2018. Plantation and Management Plan For Camp 4, Cox's Bazar. UNHCR, IUCN (International Union for Conservation of Nature), Dhaka, Bangladesh. JRP, 2020. JPR for Rohingya Humanitarian Crisis in Bangladesh.
- Wang, D., Wu, B.S., Li, F., Li, X., Wang, Z., Hou, J., Cao, R., Yang, W., 2023. Soil organic carbon stock in China's tea plantations and their great potential of carbon sequestration. J Clean Prod 421, 138485. <https://doi.org/10.1016/j.jclepro.2023.138485>.
- Zaman, M.A., Osman, K.T., Haque, S.M.S., 2010. Comparative study of some soil properties in forested and deforested areas in Cox's Bazar and Rangamati Districts, Bangladesh. J. For. Res. 21 (3), 319–322.